



Cyberster Blue Team
Internship
WEEK 11 Report

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CSI-B1-487

Technical Objective

The objective of this phase was to perform a deep-dive host and network-level digital forensic audit on a provided EWF bit-stream image

([nps-2008-jean.E01](#)) to reconstruct an unauthorized data exfiltration event.

The scope of work involves evaluating email database archives ([.pst](#)), unmasking social engineering vectors, identifying local shortcut metadata files ([.lnk](#)), and mapping system artifacts within a strict temporal timeline to establish accurate attribution and intent.

Analysis Workstation Path: [Desktop/rayyan/autopsies](#)

Forensic Suite: Autopsy Core Engine (v4.23.0)

Target Image Baseline: [nps-2008-jean.E01](#)

Part A: Email Archive Extraction & Mail Triage

In this phase of the investigation, we performed a structural validation of local email repository containers and executed a raw header routing analysis to trace deceptive communications.

1. Email Archive Inventory

To establish the baseline data store, the target user profile's local storage was audited to isolate application-layer communication databases.

- **Action Performed:** Navigated to the application data path of the default user profile and extracted the primary Outlook database properties.

- **Metrics & Findings:**

- **File Name:** [outlook.pst](#) ○ **File**

- Path:**

- [/img_nps-2008-jean.E01/vol_vol2/Documents and Settings/Jean/Local Settings/Application](#)

Data/Microsoft/Outlook

- **Exact File Size:** 2,326,528 bytes (Approx. 2.22 MB)
- **Ingestion Status:** 261 total email messages successfully parsed from the container archive.

Source Name	S	C	O	E-Mail From	Subject	Date Received	Message (Plaintext)
outlook.pst				Microsoft Outlook 2000	Welcome to Microsoft Outlook 2000!	2008-07-10 13:47:23 PKST	Welcome to Microsof
outlook.pst				Google Alerts <googlealerts-noreply@google.com>	Google Alert - skin in the office	2008-07-18 09:19:43 PKST	
outlook.pst				Google Alerts <googlealerts-noreply@google.com>	Google Alert - m57.biz	2008-07-17 00:55:25 PKST	
outlook.pst				alex <alex@m57.biz>	FW: UFOs Over U.S. Military Sites?	2008-07-20 05:32:54 PKST	CNN.com - Larry King
outlook.pst				alex <alex@m57.biz>	FW: Making People Sick AND Poor	2008-07-20 05:32:54 PKST	CNN.com - Lou Dobb
outlook.pst				alex <alex@m57.biz>	FW: Subject line: Missing girl's mom borrowed a shovel?	2008-07-20 05:32:54 PKST	CNN.com - Prime Ne
outlook.pst				alex <alex@m57.biz>	FW: The CNN Political Ticker AM for Friday, July 18, 2008	2008-07-20 05:32:53 PKST	CNN.com - CNN Polit
outlook.pst				alex <alex@m57.biz>	FW: Fans ready to stay up all 'Knight' for Batman movie	2008-07-20 05:32:52 PKST	-----Original Message
outlook.pst				alex <alex@m57.biz>	FW: All In All, I Feel Like Another Brick In the Wall	2008-07-20 05:32:52 PKST	The Morning Express
outlook.pst				alex <alison@m57.biz>	RE: which email address are you using?	2008-07-20 05:50:19 PKST	Yes, I got this email--
outlook.pst				Microsoft Outlook 2000	Welcome to Microsoft Outlook 2000!	2008-07-06 13:38:43 PKST	Welcome to Microsof
outlook.pst				Jean User <jean@m57.biz>	test test test	2008-07-06 13:48:26 PKST	Do I have email now?
outlook.pst				Jean User <jean@m57.biz>	let's try again	2008-07-06 13:48:26 PKST	Do I suck?
outlook.pst				jean@m57.biz <jean@m57.biz>	this is what I was talking about	2008-07-06 13:55:47 PKST	*Please note, the send
outlook.pst				Google Alerts <googlealerts-noreply@google.com>	Click to confirm your Google Alert	2008-07-06 13:56:41 PKST	Google received a req
outlook.pst				Google Alerts <googlealerts-noreply@google.com>	Click to confirm your Google Alert	2008-07-06 13:56:53 PKST	Google received a req
outlook.pst				accounts-noreply@google.com <accounts-noreply@...	Google Email Verification	2008-07-06 13:58:25 PKST	Welcome to Google A
outlook.pst				NPR Accounts <npr-accounts@npr.org>	Activate Your Account	2008-07-06 13:59:49 PKST	Thank you for register
outlook.pst				Google Alerts <googlealerts-noreply@google.com>	Google Alert - skin in the office	2008-07-06 14:00:35 PKST	

Name	S	C	O	Modified Time	Change Time	Access Time	Created Time	Size	Flags(Dir)	Flags(Meta)	Known	Location	MD5 Hash	SHA-256 Hash	MIME Type	Extension
[current folder]																
[parent folder]																
outlook.pst				2008-07-21 07:17:54 PKST	2008-07-21 07:17:54 PKST	2008-07-21 07:17:54 PKST	2008-07-06 13:38:42 PKST	2326528	Allocated	Allocated	unknown	/img_nps-2008-jean.E01/vol_vol2/Documents and Settings/jean/Local Settings/Application Data/Microsoft/Outlook	8c862a8c7ad8b7aff1df4d44fbf1fe95	e17b8c393fc3478c9a6bd59628d551798c4b...	application/vnd.ms-outlook-pst	pst

2. Raw Header Analysis & Spoofing Detection

The mail triage focused on identifying out-of-band transmission requests sent during the target incident window on July 20, 2008.

- **Action Performed:** Isolated the conversation thread containing high-urgency language and evaluated the raw RFC 2822 message headers within the Autopsy text metadata viewer.
- **Metrics & Findings:**
 - **Target Thread Subject:** "RE: which email address are you using?"
 - **Deceptive Interface Fields:** The user-facing mail client interface displayed the sender's identity as corporate leadership: alex <alison@m57.biz> or "Alison Smith".
 - **The Smoking Gun (Server Routing Data):** Inspection of the raw transport headers revealed explicit display name spoofing. The underlying mail server metadata routed all replies and attachments away from the internal corporate domain directly to an unauthorized external account: <tuckgorge@gmail.com>.
 - **Analytical Summary:** The forensic evidence confirms that the user was targeted by a deceptive spear-phishing attack. The adversary successfully exploited user-interface presentation layer trust boundaries to execute an data exfiltration trap.

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Name	S	C	O	Modified Time	Change Time	Access Time	Created Time	Size	Flags(Dir)	Flags(Meta)
LightBlueTop.Ink			0	2008-07-06 13:54:08 PKST	2008-07-06 13:54:08 PKST	2008-07-13 05:35:46 PKST	2008-07-06 13:50:42 PKST	711	Allocated	Allocated
m57biz.Ink			0	2008-07-20 07:28:04 PKST	2008-07-20 07:28:04 PKST	2008-07-20 07:28:04 PKST	2008-07-20 07:28:04 PKST	468	Allocated	Allocated
My Pictures.Ink			0	2008-07-06 13:54:08 PKST	2008-07-06 13:54:08 PKST	2008-07-13 05:35:46 PKST	2008-07-06 13:50:42 PKST	446	Allocated	Allocated

Save Table as CSV

Hex Text Application File Metadata OS Account Data Artifacts Analysis Results Context Annotations Other Occurrences

Metadata

Name: /img_nps-2008-jean.E01/vol2/Documents and Settings/Jean/Recent/m57biz.Ink
Type: File System
MIME Type: application/octet-stream
Size: 468
File Name Allocation: Allocated
Metadata Allocation: Allocated
Modified: 2008-07-20 07:28:04 PKST
Accessed: 2008-07-20 07:28:04 PKST
Created: 2008-07-20 07:28:04 PKST
Changed: 2008-07-20 07:28:04 PKST
MD5: 267c4ad8a74c278fa0d5013342b43b64
SHA-256: 862a9e36e9ae10ee821817bf96b65ca04b966b20455bc20f61b8a334730825d8
Hash Lookup Results: UNKNOWN
Internal ID: 12828

From The Sleuth Kit istsat Tool:
MFT Entry Header Values:
Entry: 32704 Sequence: 3

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Part B: System Artifacts & Behavior Reconstruction

Part B shifts the focus from network-layer email routing metadata to local host actions to track the exact lifecycle and execution path of the compromised document.

1. Local File Provenance & Execution Context

Windows system shortcut caches were exhumed to determine how and where the target spreadsheet was handled on the endpoint.

- **Action Performed:** Navigated to the user profile's shortcut history cache directory and extracted parsed Master File Table (MFT) records using the Sleuth Kit `istat` metadata layer.
- **Metrics & Findings:**
 - **Shortcut File Name:** `m57biz.lnk` ○
Shortcut File Path:
`/img_nps-2008-jean.E01/vol_vol2/Documents and Settings/Jean/Recent/m57biz.lnk`
 - **Shortcut Size:** `468 bytes`
 - **MFT Timestamp (\$STANDARD_INFORMATION):** `2008-07-20 01:28:04 UTC` (Displayed locally as `07:28:04 PKST`).
 - **Internal Target Path String:** Decoding the internal `.lnk` stream structures extracted the absolute local path of the spreadsheet:
`C:\Documents and Settings\Jean\Desktop\m57biz.xls` (with a relative link path pointing to `..\Desktop\m57biz.xls`).
 - **Analytical Summary:** This establishes that the spreadsheet file was actively opened and manipulated directly within the user's desktop workspace space at exactly `01:28:04 UTC`. This aligns perfectly with the target outbound exfiltration timeline window.

Save Table as CSV										
Name	S	C	O	Modified Time	Change Time	Access Time	Created Time	Size	Flags(Dir)	Flags(Meta)
LightBlueTop.Ink			0	2008-07-06 13:54:08 PKST	2008-07-06 13:54:08 PKST	2008-07-13 05:35:46 PKST	2008-07-06 13:50:42 PKST	711	Allocated	Allocated
m57biz.Ink			0	2008-07-20 07:28:04 PKST	2008-07-20 07:28:04 PKST	2008-07-20 07:28:04 PKST	2008-07-20 07:28:04 PKST	468	Allocated	Allocated
M...Picture.Ink			0	2008-07-06 13:54:08 PKST	2008-07-06 13:54:08 PKST	2008-07-13 05:35:46 PKST	2008-07-06 13:50:42 PKST	446	Allocated	Allocated

Hex	Text	Application	File Metadata	OS Account	Data Artifacts	Analysis Results	Context	Annotations	Other Occurrences
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Metadata	
Name:	/img_nps-2008-jean.E01/vol_vol2/Documents and Settings/Jean/Recent/m57biz.Ink
Type:	File System
MIME Type:	application/octet-stream
Size:	468
File Name Allocation:	Allocated
Metadata Allocation:	Allocated
Modified:	2008-07-20 07:28:04 PKST
Accessed:	2008-07-20 07:28:04 PKST
Created:	2008-07-20 07:28:04 PKST
Changed:	2008-07-20 07:28:04 PKST
MD5:	267c4ad8a74c278fa0d5013342b43b64
SHA-256:	862a9e36e9ae10ee821817bf96b65ca04b966b20455bc20f61b8a334730825d8
Hash Lookup Results:	UNKNOWN
Internal ID:	12828

From The Sleuth Kit istat tool:	
MFT Entry Header Values:	
Entry:	32704 Sequence: 3

Part C: Final Comprehensive Forensic Timeline

To complete the triage, a high-signal chronological master timeline was synthesized across a strict two-minute window surrounding the event (01:27:00 UTC to 01:29:00 UTC on July 20, 2008) to isolate concurrent system changes, browser interaction, and execution layers.

Date / Timestamp (UTC)	Artifact Source	Analytical Observation / Event Details
2008-07-20 01:28:03	File System Allocation	Modification event recorded for the desktop file directory cluster pointer (/Documents and Settings/Jean/Desktop).

2008-07-20 01:28:03	System Executable Link	Structural modification logged for /WINDOWS/system32/mf3216.dll .
2008-07-20 01:28:04	Windows Shortcut (.lnk)	Execution trigger creates m57biz.lnk pointing directly to source file C:\Documents and Settings\Jean\Desktop\m57biz.xls .
2008-07-20 01:28:04	Internet Explorer History	Write/creation entry logged inside the IE5 history database engine file: .../History.IE5/MSHist012008072020080721/index.dat , tracking an active link update.
2008-07-20 01:28:04	System Volume Information	Active OS checkpoint update written to the configuration log: .../_restore{72FA17C4...}/RP22/change.log.2 .
2008-07-20 01:28:13	Application Execution	Microsoft Excel process initialization completes; system executes application installer dependency asset xlicons.exe and modifies Excel.pip (Excel Personalized Menu Settings).

Final Forensic Case Conclusion

The comprehensive host and network-layer forensic audit conclusively exonerates Jean of acting as a malicious insider threat. The parsed data paths demonstrate a classic, highly effective spear-phishing attack.

The threat actor exploited corporate leadership's identities via display name spoofing to manufacture extreme situational urgency. The system timeline logs confirm that the user opened the file locally ([m57biz.xls](#)) and responded to what she believed was an internal request, while

transport headers secretly redirected the outbound mail path to route the proprietary corporate spreadsheet directly to an external unauthorized actor at tuckgorge@gmail.com.